

REPO: harness/harness

GitHub: <https://github.com/harness/harness>

Type: Open-source CI/CD + Git Hosting Platform

Runtime: Docker (self-hosted), localhost:3000

1. What It Is

Harness (open-source edition) adalah platform CI/CD + Git hosting yang bisa di-self-host. It combines a Gitea-compatible Git server, pipeline runner, and web UI into a single Docker deployment. Originally known as Drone CI, it has evolved into a full platform that handles code hosting, automated testing, build pipelines, and deployment triggers without requiring GitHub Actions or external CI services.

2. Tech Stack

- **Backend:** Go
 - **Frontend:** React
 - **Runtime:** Docker Compose — `single docker -compose up` starts Git server + CI runner + UI
 - **Database:** SQLite (default, good for solo use) or PostgreSQL
 - **Port:** 3000 (web UI + Git), configurable
 - **Requirements:** Docker >= 24, 4 GB RAM minimum, port 3000 available
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3. Kegunaan untuk ResistanceZero

NOW:

- Automate full deploy pipeline: push ke Harness Git → pipeline trigger → `task minify` → `git push origin main` ke GitHub Pages
- Replace manual build steps: setiap commit ke Harness repo otomatis run linting, minification, dan link checking
- Mirror rz-work repo ke local Harness instance sebagai backup + CI layer

FUTURE:

- **Obsidian sync pipeline:** vault update → Harness detects change → trigger WeKnora re-index
 - **PLN dataset quarterly refresh:** pipeline terjadwal yang run `tools/build-osm-dataset.py` setiap Jan/Apr/Jul/Oct (complement ke scheduled routine yang sudah ada)
 - **Multi-site deploy:** satu pipeline yang build Dunia-Emosi, rz-work, dan dcmoc secara paralel setelah merge ke main
 - **Artifact storage:** simpan minified JS/CSS sebagai pipeline artifacts dengan versioning
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4. Installation

```
# Create config directory
mkdir -p ~/Apps/harness && cd ~/Apps/harness
# Get docker-compose file
```

```
curl -O https://raw.githubusercontent.com/harness/harness/main/docker-compose.yml
# Configure ports and volumes (optional: edit docker-compose.yml)
# Default: port 3000, SQLite database
# Start Harness
docker compose up -d
# Access web UI
xdg-open http://localhost:3000
# Initial setup: create admin account via web UI wizard
# Register your rz-work repo as a new repository
```

5. Integrasi

Tool rz-work	Cara Integrasi
Taskfile (go-task)	Pipeline steps call `task minify`, `task deploy`, etc. – Taskfile adalah canonical build interface
Python tools	`tools/build-osm-dataset.py` dijalankan sebagai pipeline step di Harness runner
GitHub Pages	Harness pipeline push ke `baguspermana7-cpu/bfrfranco.github.io` via git push dengan deploy key
WeKnora	Pipeline trigger: setelah vault files berubah, call WeKnora ingest API
`pln-osm-quarterly-refresh` cron	Harness bisa duplicate scheduled trigger ini sebagai fallback atau primary scheduler

Pipeline YAML example location: ~/Apps/harness/pipelines/rz-deploy.yml

6. Contoh Penggunaan

```
# .harness/pipeline.yml – rz-work deploy pipeline
pipeline:
  name: rz-deploy
  identifier: rz_deploy
  stages:
    - stage:
      name: Build & Minify
      type: CI
      spec:
      steps:
        - step:
          name: Install deps
          type: Run
          spec:
          command: |
            npm install -g terser clean-css-cli
        - step:
          name: Minify assets
          type: Run
          spec:
          command: task minify
        - step:
          name: Deploy to GitHub Pages
          type: Run
```

```
spec:
command: task deploy
env:
GITHUB_TOKEN: <+secrets.getValue("github_token")>

# Trigger pipeline via API
curl -X POST http://localhost:3000/api/v1/repos/rz-work/hooks/trigger \
-H "Authorization: Bearer $HARNESS_TOKEN" \
-d '{"event": "push"}'
```

7. Batasan & Risiko

- **Resource heavy:** Harness Docker stack membutuhkan ~2–4 GB RAM secara idle. Di mesin yang sama dengan WeKnora + NemoClaw, bisa terjadi resource contention. Monitor dengan `docker stats`.
- **Overkill untuk solo project:** Untuk satu developer dengan manual deploy workflow yang sudah berjalan, ROI dari Harness baru terasa jika ada 3+ automated pipelines aktif. Pertimbangkan Taskfile saja untuk kebutuhan dasar.
- **Port 3000 conflict:** dcmoc Next.js dev server default juga port 3000. Salah satu harus diganti — Harness lebih mudah dikonfigurasi ulang.
- **Git hosting redundancy:** Jika sudah pakai GitHub sebagai canonical remote, menambah Harness Git server menambah satu lagi place to sync. Pertimbangkan Harness hanya sebagai CI runner (tanpa Git hosting).
- **Docker dependency:** Jika Docker daemon crash, seluruh CI pipeline berhenti. Bukan masalah untuk dev machine, tapi perlu diingat.